

Principles of sports injury rehabilitation

with HÅVARD MOKSNES and PHIL GLASGOW

The pearl of long-term rehabilitation is to build a small team of core people who collaborate with other professions. Never walk alone!

Kjetil Jansrud, alpine skiing gold medallist at the 2014 Sochi Winter Olympic Games, 13 months after rupturing his anterior cruciate ligament

Sports injury rehabilitation is a dynamic, structured process that aims to:

- restore the injured athlete's function and performance level
- return the athlete to sports participation in a safe and timely manner
- minimise the risk of re-injury.

As previous injury is a prominent risk factor for future injury,¹⁻³ rehabilitation is a critical aspect of our professions. In most cases, the goal should be to improve the athlete's physical function to above their pre-injury level.

The foundation of sports injury rehabilitation is a targeted exercise program that is progressed gradually. As outlined in previous chapters (including 6 and 17), exercise therapy acts at the local tissue level and in the central nervous system. It may be used as a direct injury treatment (mechanotherapy)⁴ or to unload injured tissue via altered movement and muscle activation patterns. It is also important to maintain the athlete's condition as much as possible throughout the rehabilitation process. A model of the parallel priorities of exercise prescription during the rehabilitation process is shown in Figure 18.1.

Exercise prescription, progress and supervision is often performed by a physiotherapist (or similar health professional, for example sports rehabilitator, trainer, therapist). However, where possible, a broader multidisciplinary team including the sport and exercise medicine physician, orthopaedic surgeon and other sport scientists (such as strength and conditioning specialists)

should collaborate in rehabilitation planning. It is also critical to engage coaches in all phases of rehabilitation. In the early phases, coaches need to understand the plan and appreciate the functional milestones. In the later phases they should take an increasingly active role in implementing the program (Fig. 18.1).

Active rehabilitation is often supplemented with medical and manual therapies that may enhance the effects of exercise through pain management and improved tissue adaptations (Chapter 17). The success of rehabilitation depends on introducing the most effective intervention at the right time in an adequate dosage.⁵

Modern sports injury rehabilitation is progressed through phases based on sound clinical reasoning, sequenced functional achievements and the completion of functional milestones. At the same time, knowledge of tissue-specific biological healing processes should be respected and will guide the rehabilitation timeline. Exercise prescription, communication and clinical reasoning are core skills for clinicians involved in rehabilitation of sports injuries. Although most experienced clinicians are probably subconsciously following a progression model through rehabilitation, few theoretical models have been published.⁵

This chapter outlines the general principles of sports injury rehabilitation and provides a framework for progression through phases.

GENERAL PRINCIPLES

The primary aim of rehabilitation is to return the athlete to sport with appropriate function and fitness to participate at the desired level with a low risk of re-injury.

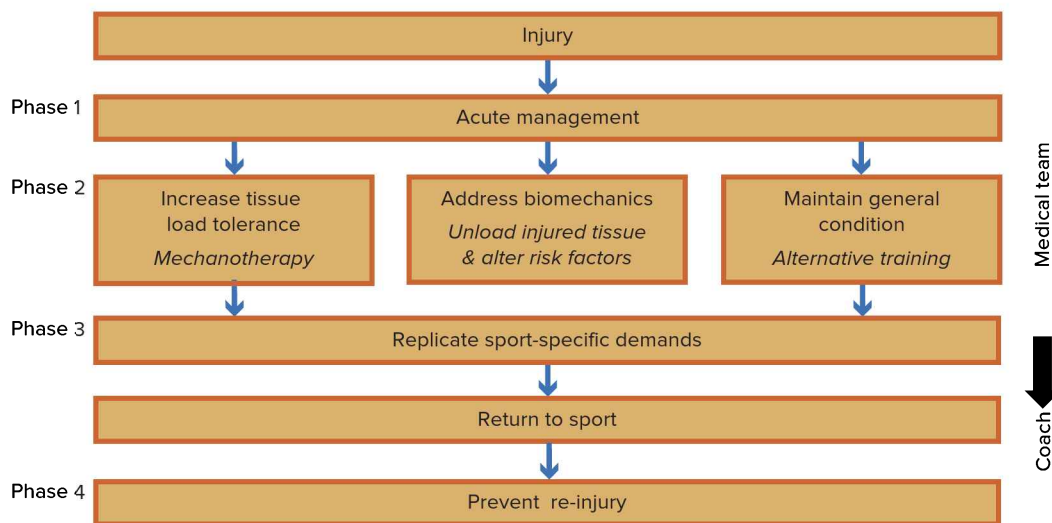


Figure 18.1 A model of the parallel priorities of exercise prescription during sports injury rehabilitation. The responsibility for implementation of the program should be gradually transferred from the medical team to the coach throughout the later phases of rehabilitation

However, this is sometimes challenging due to the desire of the athlete, coach, parents, management and others to return athletes to competition prematurely. Regardless of the circumstance, the medical team is obliged to ensure that the athlete's long-term health is not compromised by a premature return to sport which may have serious consequences. Thus, all rehabilitation protocols should include a predefined sport-specific functional battery of tests to ensure that the athlete is fit to return to sport with minimal risk of re-injury.

The rehabilitation principles described in this chapter primarily apply to the treatment of acute sports injuries and post-operative musculoskeletal rehabilitation. These rehabilitation processes are usually relatively linear and, for the most part, predictable. The principles can also be applied to overuse injuries. However, because the symptoms of overuse injuries often fluctuate and athletes often continue to train and compete with pain/impaired function, athletes may present at different stages of the rehabilitation continuum.

Rehabilitation plans following sports injury are usually separated into distinct phases and include active interventions aimed at addressing body impairments and functional limitations, with the aim of facilitating the athlete's participation in his or her desired physical activity and sport.⁶

An essential element—effective planning

Effective detailed planning is one of the most important aspects of rehabilitation. The plan should integrate

accurate anatomical, pathophysiological, biomechanical and sport-specific knowledge with physical training principles.

PRACTICE PEARL

When designing a rehabilitation program, the clinician should identify specifically how the injury affects or impairs the athlete's function—taking into account the sport and the player's role within it (e.g. their event, discipline or position).

For example, a professional athlete can devote more time to rehabilitation and to rest between exercise bouts than an amateur athlete with a demanding work schedule. This will influence the total volume of training that can be tolerated. Top athletes with day-to-day follow-up can usually progress more rapidly than recreational athletes who have less frequent encounters with their clinician. Hence, the clinician has to create a larger safety buffer when designing rehabilitation programs for non-professionals. Furthermore, physiological responses to rehabilitation stimuli vary between individuals, making the prediction of recuperation following an exercise more uncertain. Thus, the expected time to recovery differs depending on the injury, intrinsic personal and contextual factors, and the amount of stimuli that can be applied to facilitate tissue healing and recovery.

GOAL SETTING AND TARGETED INTERVENTIONS

Early goal setting and an outline of the rehabilitation plan are important elements that should be addressed in collaboration with the athlete, coach, physiotherapist and other members of the clinical team.⁷ In the first meeting between the medical team and the athlete, the key steps should include:

1. obtaining a thorough history, including the current injury, previous injuries and available resources
2. setting long-term goals
3. striving for an accurate diagnosis, which may be provisional and subject to supplementary investigations
4. identifying key impairments in body structure and function
5. raising potential physiological, psychological and logistical barriers
6. drafting a realistic progress plan that includes functional and measurable short-term goals.

Identifying the athlete's ultimate goal is usually straightforward—most athletes want to return to participation in sport as quickly as possible. However, the expectations of a rapid return may sometimes be unrealistic, and misconceptions can undermine the collaboration and focus of the rehabilitation. Thus, giving the athlete an early outline of the expected progress with specific short-term goals accompanied by specific tasks will usually build a bond of confidence. Communicating the expected progress plan to coaches, team members, family and sometimes the media usually fosters a collaborative environment for focused rehabilitation.

Physical impairments and limitations are commonly subdivided into the areas of restricted range of motion (ROM), motor control and muscle strength. Different injuries will, in combination with previous injuries and fitness status, result in different impairments and limitations (Fig 18.2). The medical team should analyse the status of the injury and decide to what extent each of the impairments or limitations should be targeted during rehabilitation. The skill of prioritising exercise interventions throughout the rehabilitation has been designated as the 'X factor' in Figure 18.2. Prioritising rehabilitation interventions relies on the ability of the clinician and the broader team to effectively synthesise complex information (via skilled clinical assessment) and identify key limitations to performance as well as barriers to progression. While a large part of the 'X factor' will be based on sound clinical reasoning, it also involves the clinician's ability to relate to the athlete, to collaborate with other professions, to motivate and guide the athlete, and to educate key persons.

PRACTICE PEARL

The clinician must clearly indicate that the rehabilitation plan is dynamic—the focus will change depending on the patient's response to the program.

Endurance and its specific subcomponents may also have to be integrated throughout the phases of rehabilitation. Examples are strength endurance for alpine skiers and speed endurance for footballers. The ability to

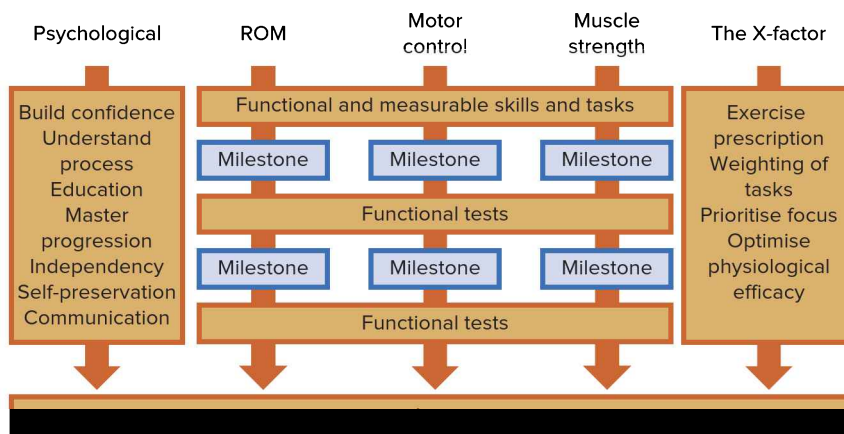


Figure 18.2 A theoretical model of clinical focus throughout the sports injury rehabilitation process

continue to carry out a specific task or exercise for an extended period of time without a reduction in the quality or effort of the task is also an important component of the rehabilitation process. Fatigue has been identified as a risk factor for injury,⁸ therefore, ensuring that the athlete is able to maintain good technique under fatigue is essential. Particular attention should be given to enhancing specific sub-qualities of endurance, such as skill execution under fatigue, as well as other strength and mobility parameters.

PHASES OF REHABILITATION

A four-phase structure can guide the aims and content of the rehabilitation process (Fig. 18.3). Specific functional milestones and achievement goals are identified within each phase. Some goals will be primary in each phase; for example, achieving full knee extension and quadriceps activation after knee surgery in Phase 1. At the same time, preservation of gluteal muscle activation and ankle mobility will be important secondary areas of focus. Therapists with different backgrounds, knowledge and experience will choose different exercises and treatments to achieve rehabilitation aims. Although there is now consensus regarding the importance of individually tailored rehabilitation after injuries,⁹ well-documented rehabilitation guidelines are scarce for most conditions in sports medicine. Also, treatment algorithms differ between cultures, countries and institutions.

Due to the multidimensional nature of sports rehabilitation, the clinician should be able to manage several parallel focus areas throughout the various phases of rehabilitation, each with different goals and milestones. The relative importance of each focus area will change throughout the rehabilitation process (Fig. 18.3). Progressing active rehabilitation requires continuous evaluation, which is necessary to guarantee a confident and timely return to training and competition.

Phase 1: Acute

In line with the increased focus on active rehabilitation strategies, the traditional acronym PRICE (Protection, Rest, Ice, Compression and Elevation) for treatment of acute soft tissue injuries is increasingly replaced with POLICE (Protection, Optimal Loading, Ice, Compression and Elevation).¹⁰ Optimal loading refers to the clinician evaluating the ability of the tissue to withstand and adapt to mechanical load.¹¹ The clinical challenge is not only to choose the most effective and safe exercise interventions, but also to apply the intervention in the appropriate dose. Immediate protection and relative rest after an acute injury still plays a role, but because of the detrimental effect of inactivity on biological tissues, leading clinicians look to have the athlete begin active movement (optimal loading) as soon as possible. For example, a player may begin hamstring activation exercises (loading) within 24 hours of a hamstring muscle strain depending on the severity of the injury and other factors.

Goal setting in the acute phase should be unidimensional and targeted toward impairments and limitations. For example, after knee ligament surgery the acute phase goals are for the athlete to regain voluntary quadriceps activation toward end-range extension (impairment) and to perform a straight-leg raise without extension lag (limitation). The combination of these two comprises the functional milestone of controlled active terminal knee extension, which allows the athlete to progress to walking without crutches for short distances.

The next goal could then be for the athlete to perform a step-up exercise with proper alignment and controlled loaded terminal extension (Fig. 18.4). When adequate step-up quality of movement is achieved, the use of crutches can be terminated for all distances. Blanchard and Glasgow have depicted the process for how to progress single exercises and this can be adapted to the rehabilitation process in general (Fig. 18.5).⁵ The model describes how the clinician can build and manipulate the

	ROM	Motor control	Muscle strength	Exercise prescription
Phase 1	Regain active & passive range	Proprioception & motor output	Voluntary activation	Controlled
Phase 2	Cyclic, full-range loading	Joint stability	Hypertrophy	Extrinsic stimuli
Phase 3	Manual therapies If necessary	Speed & agility	Generate strength	Complex movements
Phase 4	Maintenance & recovery	Ingrain new patterns	Power & endurance	Sport-specific

Figure 18.3 Model of rehabilitation through phases. Box outlines represent primary (green), secondary (orange) and tertiary (red) focus



Figure 18.4 The ability to perform a high step-up with good control of terminal knee extension is a common functional goal during Phase 1 rehabilitation of acute knee injuries

exercises to gradually increase the demands on the athlete, while at the same time allow the healing tissue to adapt to the increase in mechanical load.

A variety of interventions are available to the clinician in Phase 1. Traditional low-load controlled exercises are at the core of interventions. However, adjunct modalities such as neuromuscular electrical stimulation (NMES) can help the athlete regain voluntary muscular control.^{12, 13} Furthermore, manual therapies can be effective in the recovery of impairments during this phase (Chapter 17).

Phase 2: Restore activities of daily living

The aim of the second phase of rehabilitation is to allow the athlete to return to daily activities and basic sport-specific technical movements. The clinician should gradually increase the complexity of movements from single joint controlled actions to more complex tasks including movements through several biomechanical planes. For example, the overhead athlete who has regained pain-free external rotation in the scapular plane with adequate scapular control in Phase 1 will typically progress to mimic the throwing motion at low velocity in Phase 2. While the throwing motion will be the primary focus of the rehabilitation program, the clinician must ensure that secondary functions such as contralateral hip stability and ankle range of motion are maintained.

Exercises can be progressed by increasing the number of repetitions, the velocity of the movement, or the frequency (rate) of the exercises. Further, adding external loads and increasing the complexity of movement will stimulate muscle strength and motor control. Thus, Phase 2 challenges the clinician to prioritise exercises successfully. Skilled clinicians are able to keep their eye on several parallel processes at the same time (the X-factor).

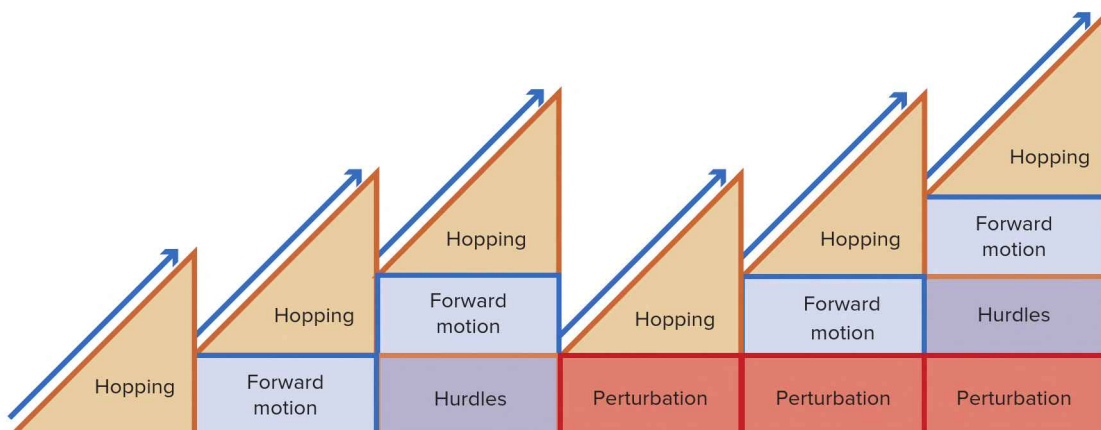


Figure 18.5 Visual aid to demonstrate a complex rehabilitation progression with undulating tiers using lower limb injury as an example⁵. The horizontal axis represents time and the vertical axis represents difficulty. The clinician adds and subtracts elements of the rehabilitation to progressively challenge the athlete while keeping the overall workload appropriate
REPRODUCED FROM FIGURE 2 IN BLANCHARD S, GLASGOW P. A THEORETICAL MODEL TO DESCRIBE PROGRESSIONS AND REGRESSIONS FOR EXERCISE REHABILITATION. *PHYS THER SPORT*. 2014 AUG;15(3):131–5

Goal setting in the second phase will still focus on the remaining impairments, although the main goals are related to addressing functional limitations in activity and improving performance of semi-complex sport-specific movement patterns. Examples of Phase 2 functional milestones for an ACL-injured athlete can be jogging with good form, and the ability to perform single leg hops with full knee extension at take-off and high degrees of knee and hip flexion during landing.

The primary intervention throughout Phase 2 is to continue progressing the therapeutic exercise program. The program will mainly incorporate elements to improve motor control and muscle strength, with additional sessions of cardiovascular training. When determining the exact nature of cardiovascular training in rehabilitation, the clinician should ensure that the correct energy system is being trained and that adequate time is allowed for recovery and physiological adaptation. For athletes who are rehabilitating over longer periods of time (i.e. months after ACL reconstruction), the acute:chronic workload ratio should be considered (Chapter 12). The nature of training used should minimise stress on the injured tissues while simultaneously exercising muscle groups involved in the athlete's sport. Aerobic and anaerobic endurance can be effectively trained using pool-based conditioning or cardiovascular equipment such as a bicycle ergometer, cross-trainer, rowing ergometer and step machine.

PRACTICE PEARL

The nature, intensity, duration and work:rest ratio of individual training sessions should reflect sporting demands.

An example of a weekly schedule may consist of two muscle strength sessions, two motor control sessions, two cardiovascular sessions and one recovery day. The precise nature of the sessions and the order in which exercises are carried out will be dictated by the clinician's understanding of the sport to which the athlete must return, the effects of the specific impairments and an appreciation of how the prescribed exercise will influence restoration of function. The clinician must have a good understanding and ability to appropriately modify load and difficulty of the exercises, while still guarding the healing tissue from abrupt forces and overload. Efficient communication with specific cues that facilitate a move from an internal focus of attention to an external focus, in combination with careful explanation and involvement of the athlete can greatly increase the efficacy and adherence to interventions.^{14, 15}

Phase 3: Returning to sports activities

In the third phase of rehabilitation the clinician guides the athlete towards returning to participation in sports. More traditional strength and conditioning training can be incorporated in the weekly rehabilitation plan with increased focus on more complexity and velocity. Emphasis should be on a higher rate of force development and introduction of on-field sessions to bring in environmental adaptation. An athlete recovering from a knee injury can be brought to the field to perform sessions focused on higher speed running, cutting and turning tasks, and the inclusion of relevant equipment (ball, racquet, skis, etc.). Physical conditioning in the gym will be similar to pre-season training for the specific sport, although a majority of exercises should still be unilaterally focused to stimulate adaptation and reverse any remaining impairments.

Goal setting in Phase 3 will relate to all levels of function. Typically, muscle strength measurements (impairment) and single-leg hop tests (activity) are compared with the goal of 90% of the uninjured side.^{16, 17} Participation goals can include completing specific parts of the team training sessions without symptoms during or after training. An important task for the clinician will be to monitor the overall load on the healing structures, which again calls for close collaboration and clear communication with coaches and strength and conditioning professionals.

Return-to-play decisions are perennially challenging and require a team approach that focuses on ensuring that function has been adequately restored. To facilitate safe return to sport, it is paramount that functional testing with valid and reliable outcome measures be performed. (See Chapter 16 for more on relevant outcome measures.) Multidimensional sets of tests that evaluate the different layers of function are best.¹⁸⁻²¹ However, the scientific backing regarding content and cut-off limits for the test batteries are still lacking for most injuries. The return-to-play process is discussed in Chapter 19.

Phase 4: Prevention of re-injury

All sports injuries with tissue disruption will render the athlete more susceptible to re-injury even though tissue structure and physical function may be restored after rehabilitation. In spite of the increased attention and knowledge on rehabilitation of sports injuries, the incidence of re-injury remains high in professional sports.^{22, 23} The classical model on the aetiology of sports injuries published by Meeuwisse in 1994 (see Fig.12.1) points to the importance of recognising internal risk factors such as previous injuries in the prevention of sports injuries.²⁴

PRACTICE PEARL

Every clinician involved in the rehabilitation of sports injuries must work diligently with the athlete, coach and conditioning professionals to continue targeted individualised prevention training after the athlete has returned to the competitive level.²⁵

Interventions should be aimed at impairments and functional limitations known to exist as a result of injury such as altered muscle activation patterns and inadequate landing and cutting strategies.²⁶⁻²⁸

WHEN REHABILITATION DOES NOT GO ACCORDING TO PLAN

Sports injury rehabilitation seems like a straightforward linear process when presented in this chapter (Fig. 18.6a). However, as every clinician has experienced, progression does not always go according to plan (Fig. 18.6b). Longstanding pain, recurrent effusion, persistent range of motion deficits and muscle inhibition are a few of the most common obstacles. What to do when progression fails?

A thorough look at the content and volume of the rehabilitation program is usually indicated. Are the interventions targeting the impairments they were intended to target? It is often prudent to follow the athlete through a full session to ensure proper technique and understanding

is present. Experienced clinicians will often state that 'less is more' and that splitting the original rehabilitation program into two parts is often a wise decision.

PRACTICE PEARL

It may be necessary to revisit the diagnosis and the functional assessment.

Symptoms tend to change over time—and not always in a predictable fashion. A step back to address the basic functional impairments is often necessary.

Sit down with the athlete and analyse the total load during the week. Is there enough time to recover between sessions? How much load is placed on the healing structures between training sessions? Are there any environmental challenges related to family, friends, facilities or motivation? Collaborate with the athlete to resolve underlying challenges. Ask open questions and probe their psychological status. A few days off rehabilitation may be a relief to many athletes.

Ask a colleague! A referral back to the physician or orthopaedic surgeon may be indicated. A second opinion from another clinician within the field can often open new doors. Additionally, a multidisciplinary meeting with all involved parties can adjust and streamline the expectations and responsibilities of each member of the rehabilitation team.

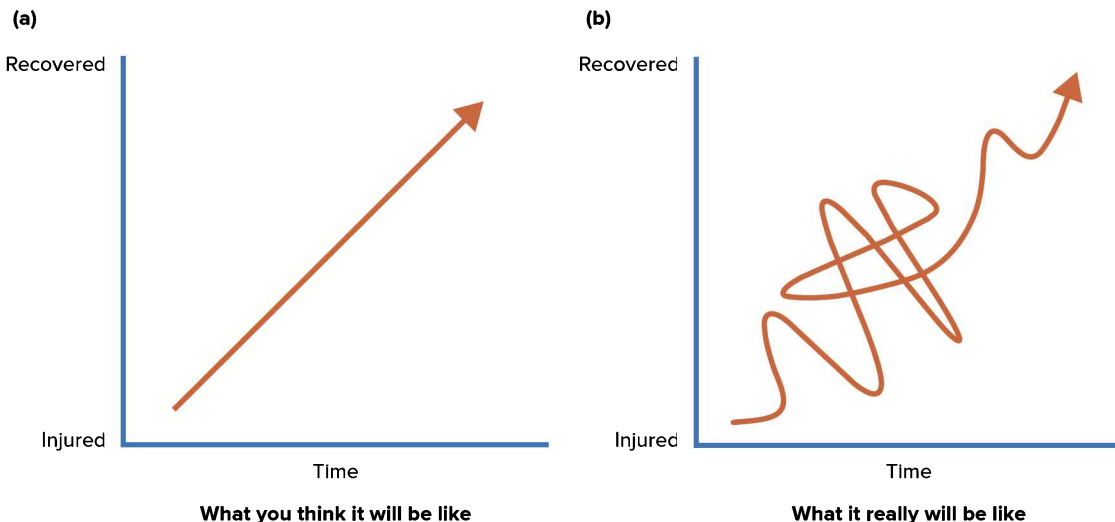


Figure 18.6 It can be helpful to share a map of the rehabilitation journey with your patient (a) Most patients believe rehabilitation should be straightforward and speedy (b) Sports physiotherapist Adam Meakins has added his own take to the traditional 'winding road'. Patients can feel they are travelling backwards in time on occasions. The expert clinician anticipates this in his or her discussion while planning rehabilitation

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e